

Week 2: Product and Category Management System

Scenario:

The pet shop wants to maintain information about the products sold in the shop and their categories. Each product belongs to a particular category, and the system should manage product details such as product name, price and stock availability.

1. Design Product and Category Tables

Category Table:

```
SQL> create table category(category_id varchar2(30), category_name varchar2(30), description varchar2(30));  
Table created.
```

Product Table:

```
SQL> create table product (product_id varchar2(30), product_name varchar2(30), category_id varchar2(30), price number(9,2), stock_quantity number);  
Table created.
```

The two tables for Category and Product have been created. Now, we have to find the relationship between the tables and apply constraints to the tables.

2. Define Relationships and Constraints

Relationships:

One Category can have many Products.

The relationship between Category and Product is One-to-Many ($1 \rightarrow *$).

We found the relationship between the tables. Now we need to identify the key that connects the two tables.

PRIMARY KEY: A column or field that uniquely identifies each row in a table.

In the Category table, Category_ID uniquely identifies each category. Similarly, Product_ID uniquely identifies each product.

FOREIGN KEY: A key that links two tables.

Therefore, Category_ID acts as a Foreign Key in the Product table.

Other Constraints:

- **NOT NULL:** Ensures that a column cannot contain an empty value. For example, a product name or category name should not be empty.
- **CHECK:** Ensures that a value satisfies a specified condition. For example, the product price must be greater than zero and stock cannot be negative.
- **UNIQUE:** Ensures that duplicate values are not allowed. For example, two categories should not have the same category name.

Defining Relationship and Creating Constraints for Product and Category:

PRIMARY KEY:

```
SQL> alter table category add constraint pk_category primary key (category_id);  
Table altered.
```

```
SQL> alter table product add constraint pk_product primary key (product_id);  
Table altered.
```

CREATING RELATIONSHIP USING FOREIGN KEY:

```
SQL> ALTER TABLE Product  
2 ADD CONSTRAINT fk_product_category  
3 FOREIGN KEY (Category_ID)  
4 REFERENCES Category(Category_ID);  
  
Table altered.
```

ADDING CONSTRAINTS:

NOT NULL:

```
SQL> alter table category modify category_name varchar(100) not null;  
Table altered.
```

```
SQL> alter table product modify product_name varchar(150) not null;  
Table altered.
```

CHECK:

```
SQL> alter table product add constraint chk_price check (price > 0);  
Table altered.  
  
SQL> alter table product add constraint chk_stock check (stock_quantity >= 0);  
Table altered.
```

UNIQUE:

```
SQL> alter table category add constraint uq_category_name unique (category_name);  
Table altered.
```

3. Insert Values

Category Table:

```
SQL> insert into category values('C001','dogs','dog related products');
1 row created.

SQL> insert into category values('C002','cats','cat related products');
1 row created.

SQL> insert into category values('C003','birds','bird related products');
1 row created.

SQL> insert into category values('C004','fish','fish related products');
1 row created.

SQL> insert into category values('C005','pet food','food for pets');
1 row created.

SQL> insert into category values('C006','pet toys','toys for pets');
1 row created.

SQL> insert into category values('C007','pet grooming','grooming products');
1 row created.

SQL> insert into category values('C008','pet accessories','accessories for pets');
1 row created.

SQL> insert into category values('C009','pet health','health care products');
1 row created.

SQL> insert into category values('C010','pet bedding','beds and bedding');
1 row created.
```

After insertion category table:

```
SQL> select * from category;
```

CATEGORY_ID	CATEGORY_NAME	DESCRIPTION
C001	dogs	dog related products
C002	cats	cat related products
C003	birds	bird related products
C004	fish	fish related products
C005	pet food	food for pets
C006	pet toys	toys for pets
C007	pet grooming	grooming products
C008	pet accessories	accessories for pets
C009	pet health	health care products
C010	pet bedding	beds and bedding

```
10 rows selected.
```

Product Table:

```
SQL> insert into product (product_id, product_name, category_id, price, stock_quantity) values ('P101','pedigree dog food','C001',550.00,25);
1 row created.

SQL> insert into product (product_id, product_name, category_id, price, stock_quantity) values ('P102','cat dry food','C002',450.00,30);
1 row created.

SQL> insert into product (product_id, product_name, category_id, price, stock_quantity) values ('P103','bird seed mix','C003',180.40,40);
1 row created.

SQL> insert into product (product_id, product_name, category_id, price, stock_quantity) values ('P104','fish food flakes','C004',120.00,35);
1 row created.

SQL> insert into product (product_id, product_name, category_id, price, stock_quantity) values ('P105','puppy chew toy','C006',250.00,25);
1 row created.

SQL> insert into product (product_id, product_name, category_id, price, stock_quantity) values ('P106','cat ball toy','C006',150.00,15);
1 row created.

SQL> insert into product (product_id, product_name, category_id, price, stock_quantity) values ('P107','pet shampoo','C007',320.00,15);
1 row created.

SQL> insert into product (product_id, product_name, category_id, price, stock_quantity) values ('P108','Dog collar','C008',280.00,18);
1 row created.

SQL> insert into product (product_id, product_name, category_id, price, stock_quantity) values ('P109','pet vitamin supplement','C009',400.00,12);
1 row created.

SQL> insert into product (product_id, product_name, category_id, price, stock_quantity) values ('P110','pet sleeping bed','C010',850.00,10);
1 row created.
```

Like this, we can insert a few more product records.

```
SQL> select * from product;
```

PRODUCT_ID	PRODUCT_NAME	CATEGORY_I	PRICE	STOCK_QUANTITY
P101	pedigree dog food	C001	550.00	25
P102	cat dry food	C002	450.00	30
P103	bird seed mix	C003	180.40	40
P104	fish food flakes	C004	120.00	35
P105	puppy chew toy	C006	250.00	25
P106	cat ball toy	C006	150.00	15
P107	pet shampoo	C007	320.00	15
P108	Dog collar	C008	280.00	18
P109	pet vitamin supplement	C009	400.00	12
P110	pet sleeping bed	C010	850.00	10

```
10 rows selected.
```

4. CRUD Operations

Insert Record:

```
SQL> insert into product (product_id, product_name, category_id, price, stock_quantity) values ('P301','pet medicines','C010',1200.00,12);  
1 row created.
```

Update Price:

```
SQL> update product set price = 3000 where product_name = 'pet medicines';  
1 row updated.
```

Delete a Product:

```
SQL> delete from product where product_name = 'pet medicines';  
1 row deleted.
```

5. Generate Category-wise Product Reports

Write SQL queries to:

- Display all products under each category.

```

SQL> SELECT c.category_name, p.product_name, p.price
2  FROM Category c
3  JOIN Product p ON c.category_id = p.category_id;

```

CATEGORY_NAME	PRODUCT_NAME	PRICE
dogs	pedigree dog food	550.00
cats	cat dry food	450.00
birds	bird seed mix	180.40
fish	fish food flakes	120.00
pet toys	puppy chew toy	250.00
pet toys	cat ball toy	150.00
pet grooming	pet shampoo	320.00
pet accessories	Dog collar	280.00
pet health	pet vitamin suppleme nt	400.00
pet bedding	pet sleeping bed	850.00

10 rows selected.

- Count the number of products in each category.

```

SQL> SELECT c.category_name, COUNT(p.product_id) AS product_count
2  FROM Category c
3  LEFT JOIN Product p ON c.category_id = p.category_id
4  GROUP BY c.category_name;

```

CATEGORY_NAME	PRODUCT_COUNT
pet bedding	1
pet food	0
fish	1
pet toys	2
dogs	1
pet health	1
cats	1
birds	1
pet grooming	1
pet accessories	1

10 rows selected.

- Find the highest-priced product in each category.

```
SQL> SELECT c.category_name, MAX(p.price) AS highest_price
2  FROM Category c
3  JOIN Product p ON c.category_id = p.category_id
4  GROUP BY c.category_name;
```

CATEGORY_NAME	HIGHEST_PRICE
pet bedding	850
fish	120
pet toys	250
dogs	550
pet health	400
cats	450
birds	180.4
pet grooming	320
pet accessories	280

9 rows selected.

- Display categories having more than 5 products.

```
SQL> SELECT c.category_name, COUNT(p.product_id) AS product_count
2  FROM Category c
3  JOIN Product p ON c.category_id = p.category_id
4  GROUP BY c.category_name
5  HAVING COUNT(p.product_id) > 5;

no rows selected
```

- Find the average product price category-wise.

```
SQL> SELECT c.category_name, AVG(p.price) AS average_price
2  FROM Category c
3  JOIN Product p ON c.category_id = p.category_id
4  GROUP BY c.category_name;
```

CATEGORY_NAME	AVERAGE_PRICE
pet bedding	850
fish	120
pet toys	200
dogs	550
pet health	400
cats	450
birds	180.4
pet grooming	320
pet accessories	280

9 rows selected.